

Stat 5200: Analysis of Designed Experiments

Jesse Wheeler

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1 Overview

Objectives and overview

- This material is based Chapter 3 on the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Chapter outcomes include:
 - TODO

2 Completely Randomized Designs

Completely Randomized Design

- Completely randomized design (CRD) is the simplest design for comparing treatments and is sometimes the preferred design.
- *Setup*:
 - There are g treatment groups.
 - N experimental units (EU)s available.
 - You want sample sizes of $n_1, n_2, \dots, n_g$, with $\sum_{i=1}^g n_i = N$.
- *Idea*: All possible sub-group composition are equally as likely.

Performing a Completely Randomized Design

- Randomly choose n_1 EU's for treatment 1.
- Randomly choose n_2 EU's for treatment 2.
- ...
- Randomly choose n_{g-1} EU's for treatment $g-1$.
- The remaining n_g EU's are assigned to treatment g .
- On a computer, this can be done by permuting $1, 2, \dots, N$, and assigning the first n_1 numbers to treatment 1, the next n_2 to treatment 2, etc.

Advantages and Disadvantages

- Advantages
 - Simple to do; all you need is one random permutation.
 - Simple to explain and understand.
 - Simple to analyze.
 - No restrictions on the sample sizes (other designs may restrict n_i).
 - Easy with missing values.
- Disadvantages
 - Not always possible or cost-effective.
 - The accuracy of the experiment depends on variability among the EUs, not just the variability among treatment groups. Thus, it may require more EUs (\$\$\$\$).

Example: Acid Rain

Wood and Bormann (1974) studied the effect of acid rain on trees. One experiment used 240 six-week-old yellow birch seedlings. These seedlings were divided into five groups of 48 *at random*, and the seedlings within each group received an acid mist treatment 6 hours a week for 17 weeks. The five treatments differed by mist pH: 4.7, 4.0, 3.3, 3.0, and 2.3. Otherwise, the seedlings were treated identically. After the 17 weeks, the seedlings were weighed, and total plant (dry) weight was taken as response.

Questions:

- What type of design is this, and can you describe why?
- What are the EUs, and how many are there?
- What was the response?
- What compromises may have been made in this case?
 - Damage may vary by pH level, plant developmental stage, plant species, etc., yet only pH level was directly addressed.
 - The study wishes to study the impact of acid-rain, but the available treatment was a synthetic acid rain, which may have a slightly different impact.
 - Total plant weight is an important response, but other responses measuring tree health may also be available.

Like many experiments, the researchers necessarily have to narrow down an enormous question to a workable experiment using artificial acid rain under controlled conditions. For any experiment, a considerable amount of non-statistical background work goes into the planning even the simplest experiment.

3 Models and Parameters

Introduction and definitions

- All formal statistical analysis is based on probability models.
- *EX*: The number of heads in ten tosses of a fair coin would be $\text{Binomial}(10, 0.5)$. Here, the model depends on 2 parameters: $N = 10$ the number of trials, $p = 0.5$ is the probability of heads.

- In an experiment, we may posit several different models for the data, all with *unknown* parameters. One of our objectives is often to describe which model is the best description of the data, and make *inference* about parameters.

Definition: Inference

Using data to learn properties of a probability distribution. In our case, using data to estimate parameters of a model.

- The models we will consider for experimental design will have two basic parts:
 - A description of the expected value, or mean, of the data. Sometimes called “model for the means”.
 - A description of how data vary around the treatment means. Sometimes called ‘model for the errors’.
- For the errors, we will generally assume that they are independent for different data values, have mean zero, and all have a common variance σ^2 .
- To perform formal tests and obtain confidence intervals, we will further assume a distribution of the errors. Generally, we assume that the errors follow a *Gaussian* (Normal) distribution.

TEST

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References

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